PERSISTENCE AI AGENT — 10 LAYER DESIGN WORKBOOK

I am building an AI admission agent that owns/manages student admission enquiries and follow-ups through its lifecycle from initial enquiry to enrolment. It remembers parent communication and student records, uses tools to send reminders and check admission status, makes decisions about follow-ups, asks a human when uncertain or approval is required, and I will prove its value by improving accuracy from the manual process to AI-assisted follow-ups.

Framing — Before Layer

Persistence AI Agent — 10 Layer Design Workbook

Spec Writing

The problem I am trying to solve is:

- After the parent brief, they say they'll check. Later they forget. So to fix this we'll send reminders through the agent.
- Adding students in the school group.
- Checklist on whether the students submitted all their certificates. To fix this we will create a checklist in the app.

The person experiencing the problem is:

- Divya mam (Admin)

My proposed AI-agent helps to:

- Follow up the parents to enroll their students.
- Send reminders.
- Add parents phone no. to the school group.
- Check whether they submitted the certificate required.

Layer 1 — Users

1.1 — Primary users

The person who directly interacts with the agent is HR — Divya ma'am. 75% of her work is regarding admissions and 25% of her work is syllabus books and marketing.

1.2 — Who is not the user but benefits from it?

Parents.

The other Layer 1 headings/answers are included in the Before Layer section above where legible. Any additional wording not visible in the source is not guessed.

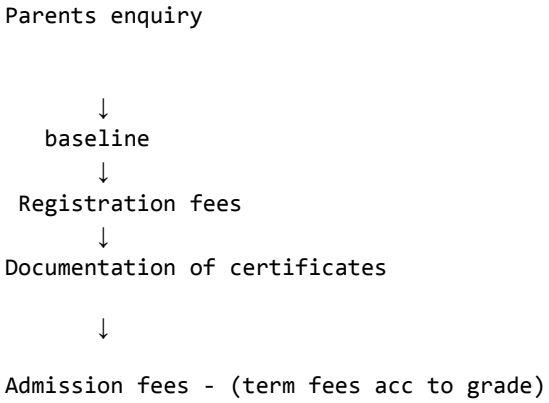
1.3 — Who pays

School / Institution

1.4 — Big frustration

"I wish I could... of the student..." [the sentence is incomplete in the source document].

1.5 — Drawing Step

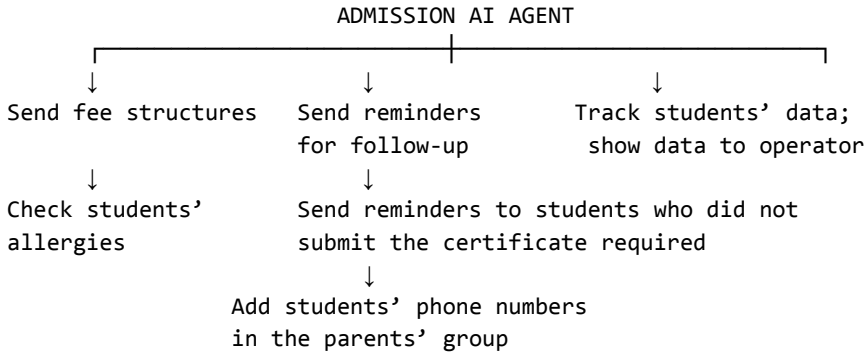


Layer 2 — Domain Entity

2.1 — Complete

The agent is responsible for managing the data of students who came to enquire for enrolling in our school and send reminders.

2.2 — Drawing steps (draw your entity)



2.3 — What information defines this entity?

S.no	Entity	Example	Where it comes from
1	Enquiry	Dr. Ram enquired for admission.	Recommendation or social media.
2	Baseline Test	Suzanna came to attend baseline test.	Question paper from admin; conducted by coordinators.
3	Certificate Submission	Dhanya submitted the required certificate with registration fee and form.	Registration form is given by Admin.
4	Fee / Admission	Jason paid his admission fees (25,000) and paid his term fees.	[Source column is crossed out / unclear in the photo.]

2.4 — What can change about the entity over time?

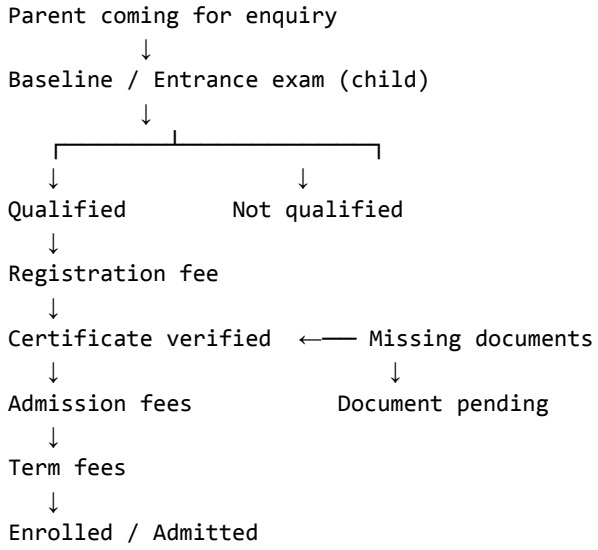
Fees structure can change over time.

2.5 — What must never be forgotten?

[The line is partly cut off in the notebook photo. Visible wording appears to refer to fee structure / follow-up and enquiry data.]

Layer 3 — State

3.1 and 3.2 — List every meaningful state



Admission AI — Transition Table 3.3

From	Event / Condition	To
Parent	Starts an admission inquiry	Waiting to Provide Student Details
Waiting to Provide Student Details	Parent submits complete and valid details	Waiting to Upload Documents
Waiting to Provide Student Details	Parent submits incomplete details	Missing Student Details
Waiting to Provide Student Details	Parent submits invalid details	Invalid Student Details
Missing Student Details	Parent provides all missing details	Waiting to Upload Documents
Invalid Student Details	Parent corrects the invalid details	Waiting to Upload Documents
Waiting to Upload Documents	Parent uploads all required documents	Waiting for Verification
Waiting to Upload Documents	Required documents are missing	Missing Documents
Missing Documents	Parent uploads the missing documents	Waiting for Verification
Waiting for Verification	Documents are valid and readable	Reviewing Admission Information
Waiting for Verification	Documents are unreadable or invalid	Invalid Documents
Invalid Documents	Parent uploads corrected documents	Waiting for Verification
Waiting for Verification	Documents contain mismatched information	Manual Review Required
Manual Review Required	Admissions officer resolves the discrepancy	Reviewing Admission Information
Reviewing Admission Information	Application meets the admission requirements	Waiting for Admission Decision
Waiting for Admission Decision	Admissions officer approves the application	Admission Approved
Waiting for Admission Decision	Admissions officer rejects the application	Admission Rejected
Waiting for Admission Decision	Applicant qualifies for the waiting list	Student Waitlisted

Student Waitlisted	A seat becomes available and the applicant is selected	Admission Approved
Admission Approved	Parent accepts the admission offer	Completing Enrollment
Admission Approved	Parent declines the admission offer	Admission Offer Declined
Completing Enrollment	Required fees are paid successfully	Enrollment Completed
Completing Enrollment	Fee payment fails	Payment Failed
Payment Failed	Parent retries and payment succeeds	Enrollment Completed
Any active state	Parent withdraws the application	Application Withdrawn
Any active state	Deadline passes without an approved extension	Application Deadline Expired
Any active state	A system error prevents processing	AI/System Error
AI/System Error	System recovers and processing resumes	Previous State

Layer 4 — Memory

4.1 What happened previously that could affect future decisions?

Previous communication history with parents, response timestamps, submission status of school transfer certificates, previous grade records, and prior counselor notes.

4.2 What decisions have humans previously made?

Manual application reviews by the principal, sibling discounts granted, age relaxation exceptions approved, or special entrance test evaluation marks set by school staff.

4.3 Should those decisions influence future cases?

Yes. If an admission coordinator grants a parent an extension to submit medical or birth certificates, the agent must remember this context to prevent sending automated threatening or overdue penalty reminders.

4.4 What should be remembered for:

- One session: Immediate question-and-answer context (e.g., curriculum details, tuition fee slabs, or grade availability inquired about by a parent).
- One entity/case: Specific student applicant details, parent contact info, uploaded report cards, and missing school admission requirements.
- One customer/user: Parent communication preferences, preferred language, and contact channel (WhatsApp vs. Email).
- Across all customers/cases: General grade-wise seat filling trends, peak admission months, and common parent roadblocks.

4.5 What should NEVER be remembered?

Raw credit card data, parent bank account details, or unmasked government identification numbers used during registration fee transactions.

4.6 Memory Architecture Flow

- Short-term memory handles active live conversations with parents via session cache.
- Case memory (Long-term) logs student candidate milestones in the school database, feeding context back into the agent whenever the parent reaches out again.

```
Parent conversation
       ↓
Short-term memory (session cache)
       ↓
```

```
Case memory / long-term school database
        ↓
Context returned when parent contacts again
```

Layer 5 — Knowledge

5.1 What documents/data does the agent need?

Knowledge	Source	Update Frequency	Classification
School Fee Structures & Payment Deadlines	School Accounts Database / Circular	Yearly	Public / Permanent
Grade Eligibility & Age Criteria	Board Guidelines / School Handbook	Per Academic Year	Public / Permanent
Frequently Asked Questions (FAQs)	School Administration Guidelines	Frequently Changing	Public / Frequently Changing
Student Application & Enrolled Status	School SIS / CRM Database	Real-time	Customer-specific / Confidential
Internal Admission Process & Escalations	School Staff SOP Document	As Needed	Company-specific / Confidential

5.2 Classify each

Public / Company-specific / Customer-specific / Confidential / Frequently changing / Permanent. (Refer to the classification tags assigned in the table above).

5.3 What happens if the agent doesn't know something?

If the data is missing from its indexed knowledge base, the agent will gracefully ask the parent for clarification or escalate to a human school admission officer while logging the question.

5.4 Knowledge Flow

```
School circulars, fee books, databases, APIs, websites, and staff
                          ↓
        Structured Vector Database / Knowledge Base
                          ↓
                  Agent retrieval
                          ↓
Contextual answers during parent inquiries
```

Layer 6 — Tools

6.1 Action Execution Table

Action	Tool / System	Read or Write?	Risk Level
Send fee or document submission reminder	WhatsApp Business API	Write	Low
Query student applicant status	School CRM Database	Read	Low
Parse and log uploaded previous report card	Document Parser API	Write	Medium
Book parent-principal interview slot	Calendar API	Write	Medium
Issue official school admission offer letter	School SIS / Email	Write	High (!)
Reject application / Waitlist student	School CRM Database	Write	High (!)

6.2 What systems must it connect to?

School CRM, Student Information System (SIS), Calendar, WhatsApp, School Website, Document Storage, and Internal Email/API.

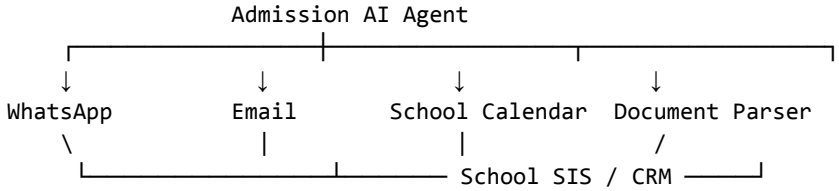
6.3 Which actions are dangerous? (Marked with !)

Sending formal acceptance/rejection notices to parents (!), updating fee concession waivers (!), or permanently modifying student seat allocation statuses (!).

6.4 Which actions require approval?

Any final decision output (such as sending official admission confirmation letters, scholarship awards, or seat allotment) must require explicit approval from the school's principal or admission head before triggering.

6.5 Agent's Tool Belt Flow



Layer 7 — Agent

7.1 What decisions require reasoning? (List)

- Answering parents' admission queries.
- Checking grade eligibility and age criteria.
- Providing fee details and deadlines.
- Deciding when to escalate queries to staff.

7.2 What information should the model receive before making each decision?

- Student's age and required grade.
- School fee structure and payment deadlines.
- Admission guidelines and seat availability.
- Parent's query and relevant school records.

7.3 What should the model NOT be allowed to decide?

- Final admission approval or rejection.
- Changes to school fees.
- Exceptions to admission rules.
- Confidential decisions reserved for school staff.

7.4 What should the agent do when uncertain? Rank: ask user / search / try another tool / ask human expert / make best judgment / stop.

1. Search the knowledge base.

2. Try another tool or database.

3. Ask the parent for clarification.

4. Ask a human expert or school staff.

5. Stop if the answer cannot be verified.

7.5 Write the agent's job description

- You are an AI: Admission Assistant.
- Your responsibility is to: Help parents with admission enquiries and provide accurate school information.
- You can: Answer questions about fees, eligibility, deadlines and admission procedures.
- You cannot: Approve or reject admissions or disclose confidential information.
- When uncertain, you: Search for verified information or ask school staff for help.
- You must ask a human when: A query requires an exception, final approval or a decision beyond your authority.

Layer 8 — Workflow

8.1 Drawing steps / workflow

```
USER
Parent asks an admission question
       ↓
AGENT – understands the question and identifies what is needed
       ↓
STATE – checks current question / case state
       ↓
TOOL – searches school database / knowledge base
       ↓
RESULT – receives information returned by the tool
       ↓
STATE CHANGE – updates the state with retrieved information
       ↓
NEXT ACTION – answers parent, uses another tool, or asks staff for help
```

Layer 9 — Human-in-the-Loop (HITL)

9.1 Confidence decision flow

```
Confidence level
       ↓
 ┌─────────────────────┴──────────────────────┐
 ↓                                            ↓
High                                         Low
 ↓                                            ↓
Execute allowed routine action               Human review
 ↓                                            ↓
Update state                                 Human decision / next action
```

The source notes describe high confidence as allowing the agent to proceed with permitted routine actions, while low confidence or decisions beyond the agent's authority require human review.

Layer 10 — Evaluation

10.1 What does good mean?

The agent must provide accurate answers and protect confidential information.

10.2 Primary metric

Primary metric: Accuracy — to measure how correctly the agent answers parents' admission questions.

10.3 Baseline

Baseline: Staff manually answer parents' admission queries without AI.

10.4 Target

Target: The AI agent answers admission queries accurately and faster, reducing staff workload.

10.5 How many real / test cases will be evaluated?

The notes discuss 10–20 test cases. For an initial evaluation, the selected answer in the later notes is: 10 test cases.

10.6 What would constitute failure?

Failures: Incorrect answers, sharing confidential data, or failing to escalate uncertain cases to staff.

10.7 Evaluation table

Case	Expected Result	Agent Result	Human Result	Correct?
Fee enquiry	Correct fee details	Correct fees	Correct fees	Yes
Age eligibility	Correct eligibility	Correct eligibility	Eligible	Yes
Admission deadline	Correct deadline	Wrong date	Correct date	No
Student status	Protect confidential data	Requests verification	Verify identity	Yes
Uncertain query	Escalate to staff	Escalate to staff	Staff handles it	yes